Creative Question (Answer any two).

1. $x^6 + 1 = 18\sqrt{2}x^3$ *and* $p^2 = 7 + 4\sqrt{3}$ *are two algebraic expressions.*

a) *Resolve into factors of* $y^2 + 2z - 1 - z^2$.
b) *Prove that* $x = \sqrt{3} + \sqrt{2}$.
c) *Determin the value of* $\frac{P^6-1}{P^3}$.

2. $(i)\ a - 2 = \frac{1}{a}$, $(ii)\ x^4 = 4(4x^2 - 1)$.
a) *Resolve into factor*: $x^2 + 4x - 21$.
b) *Find the value of* $x - \frac{1}{x}$.
c) *Find the value of* $a^5 + \frac{1}{a^5}$.

3) $(l + m)^2 = \sqrt[3]{125}$, $(l - m)^2 = \sqrt[3]{64}$ *and* $K^2 + \frac{1}{K^2} = \frac{85}{4}$.
a) *Resolve into factor*: $9c^2 + \frac{1}{9c^2} - 2 + 9c - \frac{1}{c}$.
b) *Prove that* $4(l^3m + lm^3) = \frac{9}{2}$
c) *Show that* $8(K^3 - \frac{1}{K^3}) = 89\sqrt{77}$

Short Question (Answer All).

1) *if* $a + b + c = 0$, *then prove that* $a^3 + b^3 + c^3 = 3abc$.
2) Resolve into factor: $4a^2 + \frac{1}{4a^2} - 2 + 4a - \frac{1}{a}$.
3) if $a^4 - 27a^2 + 1 = 0$, *find the value of* $a^4 - \frac{1}{a^4}$
4) *if* $x = 3 - \sqrt{8}$ *then what is the value of* $\frac{1}{x}$.
5) $p - \frac{1}{p} = 7$, what is the value of $\frac{P}{P^2-6p-1}$

Multiple Choice Question (Answer All).

1) *if* $a^2 - 4a - 1 = 0$ *what is the value of* $a^2 + \frac{1}{a^2}$.
a) 12 b) 14 c) 18 d) 20

2) *if* $a + b = 4, ab = 1$ *then* $(a + b)^2 = ?$
a) 12 b) 14 c) 18 d) 20

3) *if* $x + y = 8$ *and* $2y = 10$ *then what is the value of* x?
a) -2 b) 2 c) 3 d) 13

4) *if* $2x - y = 8$ *and* $x - 2y = 4$ *then what is the value of* $x + y$?
a) 0 b) 4 c) 8 d) 12

5) *Which one is the factors of* $x^2 - 1 + 2y - y^2$
a) $x + y - 1$ b) $x + y + 1$ c) $y - x + 1$ d) $x - y - 1$

6) *if* $\frac{1}{2-a} = 16$ *then what is the value of* a^2?
a) 4 b) 8 c) 16 d) 32

7) *if* $a^2 + \frac{1}{a^2} = 11$ *then* $(i)\ a - \frac{1}{a} = 3$, $(ii)\ \left(a + \frac{1}{a}\right)^2 = 13$, $(iii)\ a^3 - \frac{1}{a^3} = 18$
a) i & ii b) i & iii c) ii & iii d) i, ii & iii

8) *if* $a + b = \sqrt{6}$ *and* $a - b = \sqrt{5}$ *then what is value* $4ab$?
a) 1 b) $\frac{1}{2}$ c) 11 d) $\frac{11}{2}$

9) $\frac{a}{b} + \frac{b}{a} = 2$, *then* $a - b = ?$
a) 0 b) 1 c) ab d) $\sqrt{ab}$

10) $\frac{a}{b} + \frac{b}{a} = 2$ *then what is value of* $\left(\frac{a}{b}\right)^3 + \left(\frac{b}{a}\right)^3$
a) 8 b) 6 c) 4 d) 2

LESSON BY SHAFIQ